

A PROJECT REPORT
ON
VISION (Visually Impaired Smart Integrated Obstacle Navigator)

SUBMITTED IN FULFILLMENT OF REQUIREMENT FOR THE

DIPLOMA IN COMPUTER ENGINEERING

AT

GOVERNMENT POLYTECHNIC JALGAON



SUBMITTED TO

**MAHARASHTRA STATE BOARD OF TECHNICAL EDUCATION,
MUMBAI**

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CERTIFICATE

This is to certify that

- 1. Anushka Santosh Navgale**
- 2. Gargi Vijay Patil**
- 3. Jayesh Gopal Jagtap**
- 4. Omkumar Rajendra Mahajan**

of CO6K(2025-26) have successfully completed Project on “VISION”, under the guidance of “Prof. Asha Chaudhari ” in partial fulfillment of the requirement for the award of diploma in computer engineering from Maharashtra State Board of Technical Education, Mumbai at Government Polytechnic, Jalgaon

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- 1. Anushka Santosh Navgale**
- 2. Gargi Vijay Patil**
- 3. Jayesh Gopal Jagtap**
- 4. Omkumar Rajendra Mahajan**
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Date: / / **2026**

Place: Jalgaon

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ABSTRACT

Project VISION (Visually Impaired Smart Integrated Obstacle Navigator) is an advanced assistive technology solution designed to enhance the safety and mobility of visually impaired individuals. While traditional white canes offer physical feedback, they lack the capability to detect overhead obstacles, identify environmental hazards like water, or provide real-time remote tracking for caregivers.

This project implements a dual-module system comprising a Smart Navigation Stick and a Vibration/Voice-Enabled Hat, communicating via a synchronized Bluetooth Master-Slave mesh.

- The Smart Stick (ESP32 Based): Acts as the central IoT gateway. It utilizes a Neo-6M GPS module to provide real-time location tracking via the Blynk IoT platform. An integrated Emergency SOS button allows the user to instantly transmit a precise Google Maps link to a caregiver's smartphone. Furthermore, a specialized Water Sensor detects ground-level liquid hazards, triggering immediate alerts.
- The Smart Hat (Arduino Nano Based): Provides hands-free obstacle detection. Using an Ultrasonic Sensor mounted on a Servo-scanning mechanism, the hat creates a 180° "safety arc" to detect head-level obstacles. Upon detection, a DFPlayer Mini module provides localized voice feedback ("Obstacle Ahead"), while the Bluetooth link synchronizes alarms across both devices.

By integrating IoT-based remote monitoring with local sensory feedback, Project VISION offers a multi-layered safety net, bridging the gap between traditional mobility aids and modern smart technology

CHAPTER 1: INTRODUCTION

1.1 Overview

Visual impairment is a global challenge that significantly impacts an individual's quality of life, autonomy, and safety. While the traditional white cane has been the primary mobility aid for decades, it offers limited feedback, restricted primarily to ground-level physical contact. It cannot detect head-level obstacles, identify environmental hazards like water before contact, or provide a means for remote monitoring by caregivers. **Project VISION** (Visually Impaired Smart Integrated Obstacle Navigator) is developed to bridge this technological gap by integrating Internet of Things (IoT) capabilities with embedded sensor networks.

1.2 Problem Statement

The primary difficulties faced by visually impaired individuals include:

- **Undetected Overhead Hazards:** Traditional canes do not protect the user from head-level obstacles such as low-hanging branches, signboards, or open cabinet doors.
- **Environmental Uncertainties:** Detecting hazards like stagnant water or slippery surfaces often happens too late through physical contact, leading to potential accidents.
- **Lack of Emergency Connectivity:** In the event of a fall or becoming lost, there is no integrated way for the user to instantly communicate their precise location to family or medical responders.
- **Manual Dependency:** Existing methods rely heavily on the user's memory or human assistance, which is not always available.

1.3 Objectives of the Project

The core objective of this project is to create a comprehensive, low-cost safety ecosystem that provides both local and remote assistance. The specific goals include:

- **Master-Slave Synchronization:** Utilizing a master-slave Bluetooth architecture to link a Smart Stick and a Smart Hat, ensuring the user receives 360-degree protection.
- **Proactive Obstacle Scanning:** implementing a servo-driven ultrasonic "safety arc" on the hat to detect obstacles in a 180-degree field before the user reaches them.

- **IoT-Based Location Tracking:** Integrating a Neo-6M GPS module with the Blynk IoT platform to allow real-time monitoring and one-touch SOS location sharing.
- **Multi-Sensory Feedback:** Providing localized audio alerts via a DFPlayer Mini and haptic/buzzer feedback to ensure the user is immediately aware of hazards.

1.4 Innovation and Impact

Unlike standalone devices, **Project VISION** creates a network. The ESP32 acts as an IoT gateway, managing high-level tasks like cloud communication and GPS tracking, while the Arduino Nano handles local, time-sensitive sensory feedback. This distribution of tasks ensures system reliability and speed. By leveraging affordable components like the HC-SR04 and Neo-6M, this project provides a scalable solution that promotes smart agriculture principles of efficiency and resource management in the context of assistive technology.

1.5 Methodology

The project follows an iterative development cycle:

1. **Hardware Integration:** Connecting sensors and microcontrollers on a unified power rail.
2. **Firmware Development:** Writing optimized C++ code for Bluetooth pairing and GPS data parsing.
3. **Cloud Configuration:** Setting up Blynk Datastreams and Virtual Pins for mobile dashboarding.
4. **Testing and Calibration:** Real-world field testing to ensure accuracy in obstacle detection and location reporting

CHAPTER 2: LITERATURE SURVEY

Limitations of the Traditional White Cane

The traditional white cane is effective only for waist-down, single-point tactile feedback. It is inherently incapable of detecting head-level hazards (overhanging branches, signboards) or drop-offs, resulting in a 98% injury rate among users.

Emerging Electronic Travel Aids (ETAs)

Research has introduced sensor-based ETAs utilizing microcontrollers (Arduino/ESP32) with ultrasonic, IR, and moisture sensors for haptic or auditory alerts. IoT integration (Blynk, GPS) allows real-time tracking, clickable Google Maps links, and SOS notifications. advanced camera systems exist but are complex, costly, and power-intensive.

Technical Gaps in Current ETAs

1. Limited Sensing Plane: Fixed, narrow detection angles (fixed forward) fail to detect simultaneous head- and knee-level hazards.
2. Communication Latency: Singular microcontrollers managing both real-time sensing and high-latency IoT tasks can experience a long response time of 1.9 seconds.
3. Audio Ambiguity: Relying on simple buzzers/beeps rather than descriptive, localized voice prompts limits intuitive scene understanding.

Problem Statement

Current electronic travel aids for the visually impaired lack an affordable, low-latency solution providing simultaneous multi-level (head and ground) hazard detection, robust remote tracking, and distinct, intuitive voice and sweeping-frequency alerts.

CHAPTER 3 – PROJECT SCOPE AND REQUIREMENTS

3.1 The scope of **Project VISION** defines the boundaried deliverables of a low-cost electronic travel aid that synchronizes local head-level safety with remote caregiver monitoring. The functional architecture of the system is strictly limited to a standardized hierarchy:

- **In-Scope (Functional System):**
 - **Module 1: Smart Stick Master (ESP32)** handles WiFi/Cloud connectivity via the **Blynk IoT platform**, hardcoded Neo-6M GPS, SOS button alerts (debounced Pin 27), and water sensor monitoring (Analog Input). It acts as the **Bluetooth Master**.
 - **Module 2: Smart Hat Slave (Arduino Nano)** receives priority command strings (like "WATER" or "ALARM") from the Master via Bluetooth. It manages the proactive **180° safety arc** (SG90 Servo + HC-SR04 Ultrasonic). It prioritizes localized feedback: voice alerts ("Obstacle Ahead") via **DFPlayer Mini** and distinct sweeping frequency buzzer alarms (1k-3k Hz).
- **Out-of-Scope (Functional System):** Advanced computer vision features, indoor positioning systems (IPS), turn-by-turn navigation data streaming, and localized health monitoring are outside the scope of the current development phase. The system requires a stable power supply to the Arduino Nano for optimal localized servo performance.

It defines the necessary functionalities, hardware components, and software stack required to create a reliable assistive network for the visually impaired.

3.2 System Requirements

The system must operate as a synchronized network where the **Smart Stick** acts as the communication hub (Master) and the **Smart Hat** acts as the sensory extension (Slave).

3.2.1 Functional Requirements

Functional requirements define the core actions the system must perform to meet the user's needs.

- **Obstacle Detection and Scanning:** The system must use an ultrasonic sensor mounted on a servo motor to scan a 180° arc at head level.
- **Real-Time Voice Feedback:** Upon detecting an obstacle within a 50cm range, the system must trigger a specific voice alert ("Obstacle Ahead") via the DFPlayer Mini.
- **Global Positioning (GPS):** The stick must continuously acquire latitude and longitude data, by gathering live coordinates from satellite.
- **Emergency SOS Messaging:** A physical button on the stick must trigger a **Critical Event** on the Blynk IoT platform, sending a push notification and a clickable Google Maps link to the caregiver.
- **Water Hazard Sensing:** A specialized sensor at the base of the stick must detect liquid presence and send a "WATER" signal via Bluetooth to the hat for a high-frequency buzzer alarm.
- **Master-Slave Bluetooth Mesh:** The ESP32 (Master) must automatically pair with the Hat's Bluetooth address to ensure data synchronization without manual user intervention.

3.2.2 Non-Functional Requirements

These define the operational standards and quality of the system.

- **Low Latency:** The "Obstacle Ahead" voice alert must play within milliseconds of detection to ensure user safety.
- **Reliability:** The Bluetooth connection between the stick and hat must automatically reconnect if the signal is temporarily lost.
- **Power Efficiency:** The system must be optimized to run on portable battery power for extended outdoor use.
- **User Simplicity:** The interface for the visually impaired user must be entirely hands-free (voice/vibration) or limited to a single large tactile button (SOS).

3.3 Software Requirements

The software architecture integrates embedded firmware with cloud-based monitoring.

- **Development Environment:** **Arduino IDE** used for writing and uploading C++ code to both microcontrollers.

- **IoT Platform: Blynk IoT** (Mobile & Web) used for the caregiver's dashboard and push notification management.
- **Programming Language: C++** with specific libraries including TinyGPSPlus, BluetoothSerial, and DFRobotDFPlayerMini.
- **Communication Protocols:**
 - **UART (Serial):** For GPS and DFPlayer communication.
 - **Bluetooth Classic:** For the Master-Slave link between the Stick and Hat.
 - **WiFi (HTTP/SSL):** For secure data transmission from ESP32 to Blynk Cloud.
- **Cloud Tools: Blynk Web Console** for configuring Virtual Pins (V1, V2) and defining "sos_alert" event logic.

3.4 Hardware Requirements

Component	Specification/Model	Purpose
Master Controller	ESP32 DevKit V1	Handles WiFi, Bluetooth Master mode, and GPS processing.
Slave Controller	Arduino Nano	Controls the Servo, Ultrasonic sensor, and DFPlayer on the Hat.
GPS Module	Neo-6M with Antenna	Provides real-time coordinates for tracking.
Obstacle Sensor	HC-SR04 Ultrasonic	Measures distance to objects at head level.
Voice Module	DFPlayer Mini	Plays pre-recorded MP3 alerts from a MicroSD card.
Connectivity	HC-05 (for Nano)	Receives Bluetooth signals from the ESP32.
Actuators	SG90 Servo & Buzzer	Handles scanning movement and localized alarms.
Hazard Sensor	Soil Moisture/Water	Detects puddles or wet surfaces.

The project utilizes a mix of high-performance IoT controllers and specialized sensors.

CHAPTER 4 – METHODOLOGY

The methodology for **Project VISION** follows a structured engineering approach, transitioning from problem identification to a synchronized dual-module hardware implementation. The process ensures that high-latency IoT tasks (GPS/Cloud) and low-latency safety tasks (Obstacle Detection) are handled in parallel.

4.1 Risk Protection

Identifying potential technical and operational hurdles is essential for a safety-critical device like a smart navigation aid.

Risk Factor	Description	Mitigation Strategy
Connectivity Failure	The Bluetooth link between the stick and hat might drop in high-interference areas.	Implement an automatic reconnection loop in the ESP32 firmware that retries the Hat's MAC address every 10 seconds.
GPS Inaccuracy	Satellite signal loss indoors or in urban canyons could lead to incorrect location sharing.	Set GP Jalgaon (21.0034, 75.5513) as the hardcoded default location until a valid satellite lock is confirmed by the tinyGPS++ library.
Power Instability	The Servo motor and DFPlayer draw high current, which can cause the Arduino Nano to reset.	Utilize a common ground and potentially add a decoupling capacitor across the power rails to stabilize voltage during servo movement.
Audio Logic Errors	Incorrect file naming or folder structure on the SD card prevents voice feedback.	Strictly enforce the mp3 folder naming convention and use 0001.mp3 for the "Obstacle Ahead" alert.

4.2 Feasibility Study

A comprehensive evaluation confirms that the dual-module system is viable across multiple dimensions.

- **Technical Feasibility:** The use of ESP32 for IoT tasks and Arduino Nano for local sensing is highly efficient, as both support the C++ Arduino framework and offer robust library support for GPS, Bluetooth, and MP3 modules.
- **Operational Feasibility:** The device requires zero configuration from the visually impaired user. The Master-Slave pairing is hardcoded, and feedback is provided through intuitive voice and buzzer sounds.
- **Financial Feasibility:** By utilizing open-source platforms like Blynk IoT and low-cost sensors (HC-SR04, Neo-6M), the project remains affordable compared to commercial smart canes.

4.3 Behavioral and Functional Description

The system logic is divided between the Smart Stick (Command Center) and the Smart Hat (Safety Extension).

4.3.1 Behavioral Logic

1. **The Scanning Loop:** The Hat's Servo sweeps the Ultrasonic sensor between 45° and 135° to monitor the path.
2. **Detection Event:** If an object is detected within 50cm, the Servo stops, and the DFPlayer triggers the voice alert.
3. **Emergency State:** When the user presses the SOS button, the ESP32 halts background tasks to transmit the Critical Alert and GPS coordinates to the Blynk cloud.
4. **Water Detection:** Upon sensing liquid, the Stick sends a "WATER" string over Bluetooth, forcing the Hat to activate a high-frequency buzzer sweep.

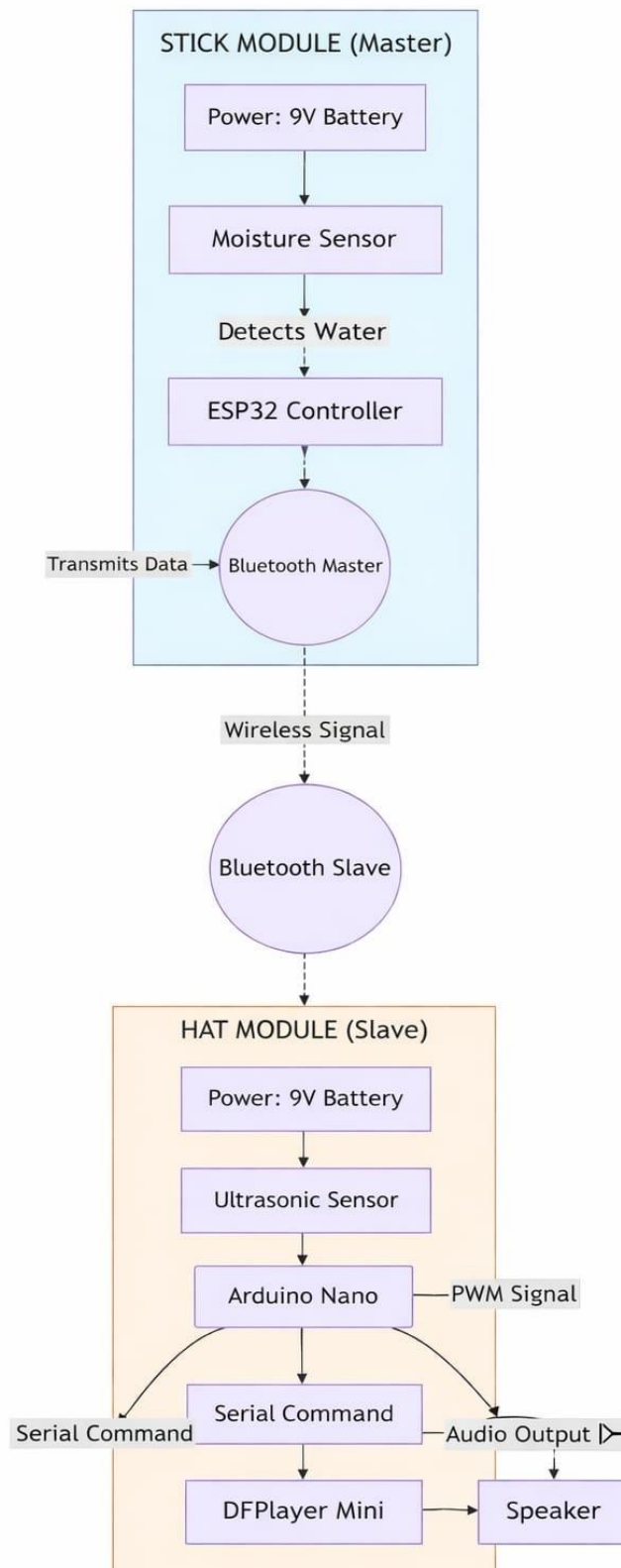
4.3.2 Functional Description

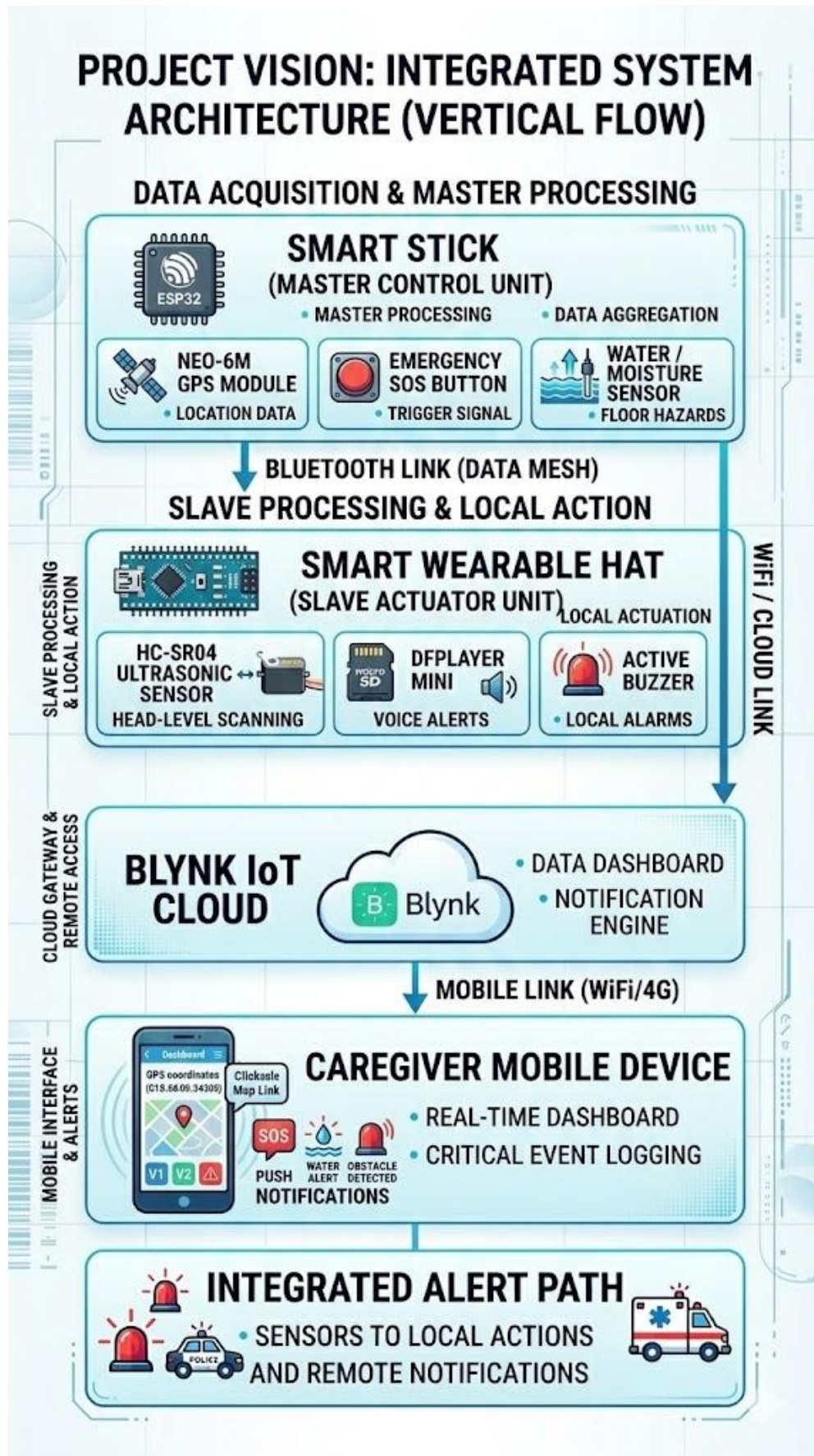
Functionality	Component Responsible	Description
Remote Monitoring	Blynk IoT & ESP32	Provides a real-time dashboard for caregivers to track the user's location.
Obstacle Feedback	DFPlayer & Nano	Converts digital distance data into human-understandable voice alerts.
Location Sharing	Neo-6M GPS	Generates a Google Maps link for emergency responders.
Bluetooth Mesh	BTSerial (Master-Slave)	Ensures the Stick can remotely trigger alarms on the Hat for integrated safety.

CHAPTER 5 - DETAILS OF DESIGN WORKING AND PROCESS

5.1 Data Flow Diagrams (DFD)

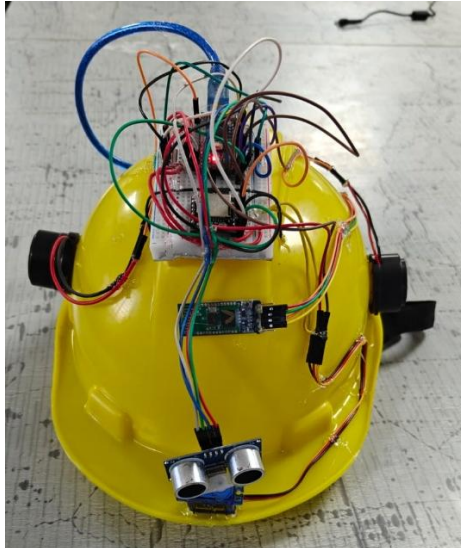
The DFDs represent the movement of data from sensors to the end-user and caregiver.



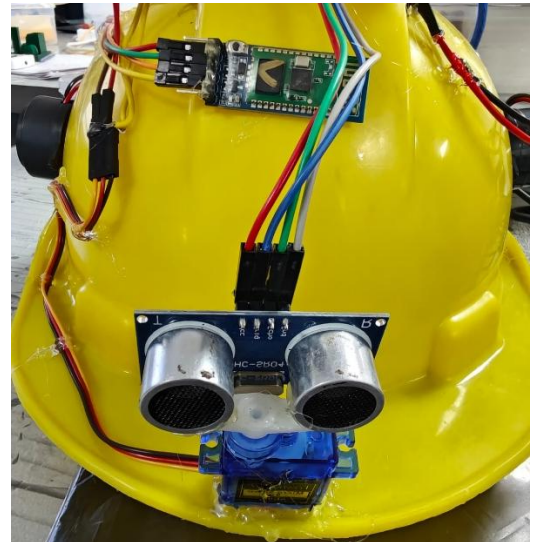


5.2 ACTUAL SYSTEM

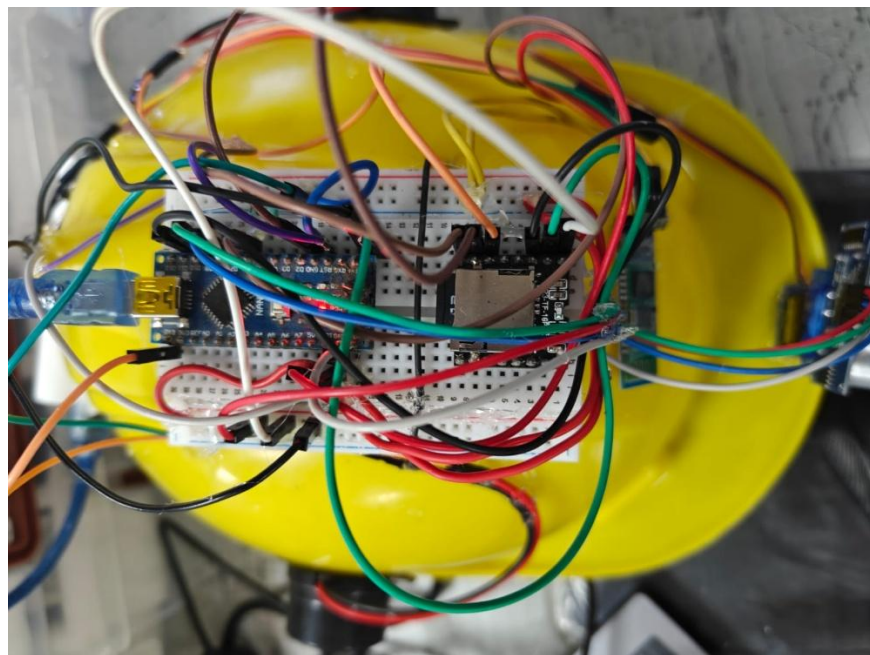
Hat Module



Ultrasonic Sensor and Bluetooth Module on Hat.



Microcontroller Ardiuno Nano and DFplayer Mini



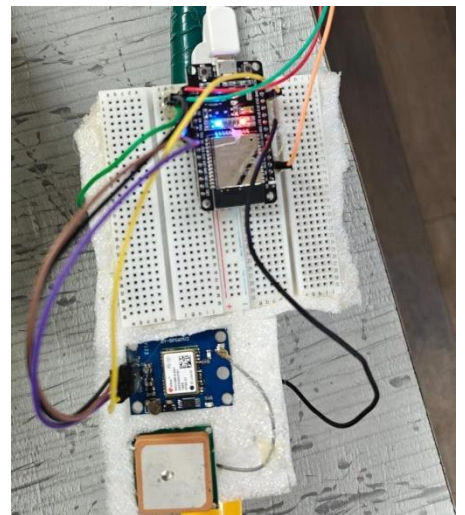
Stick Module



Emergency Push Button on Stick Module



GPS Module and Microcontroller (ESP32)

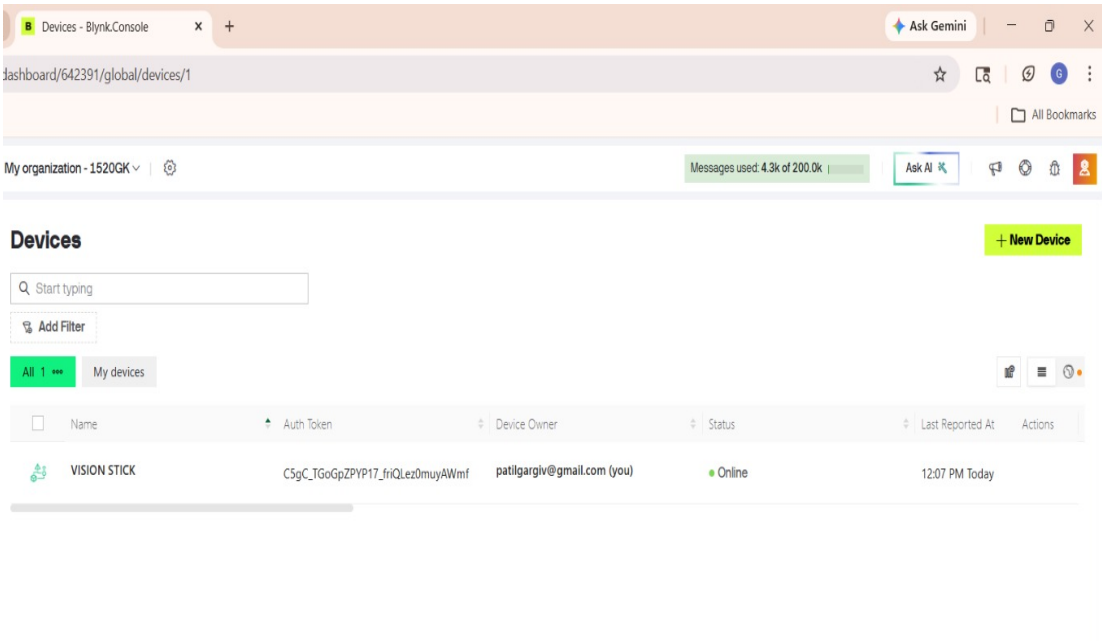


Moisture Sensor to detect puddles and Slippery Surfaces

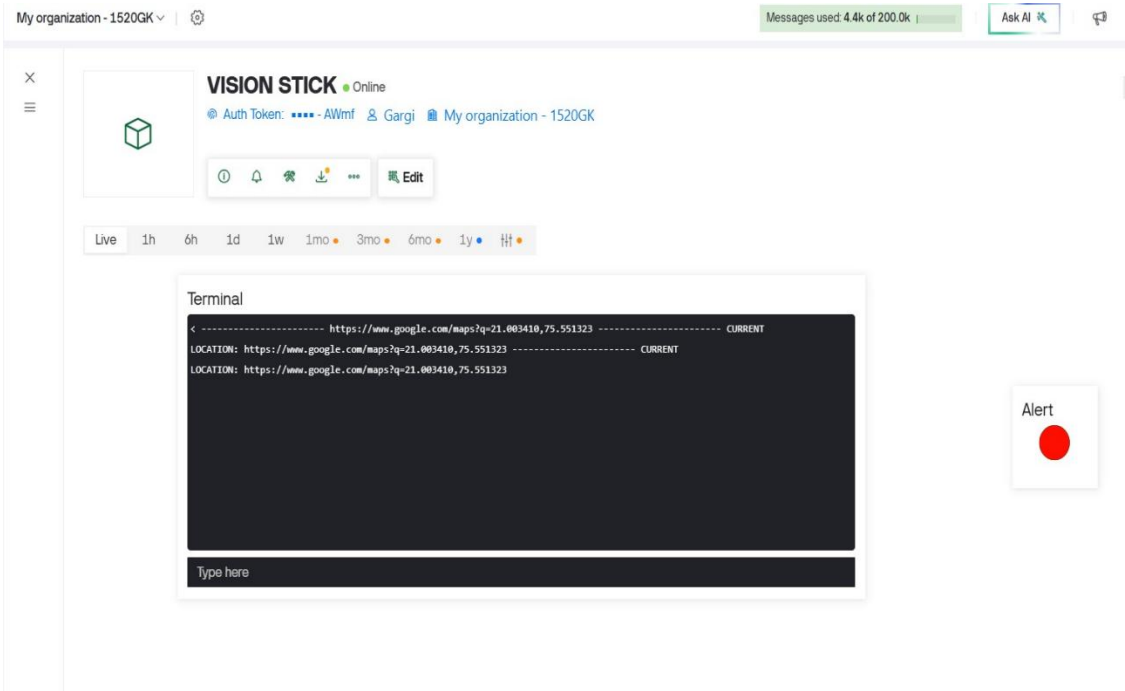


Cloud Based Application Blynk IOT

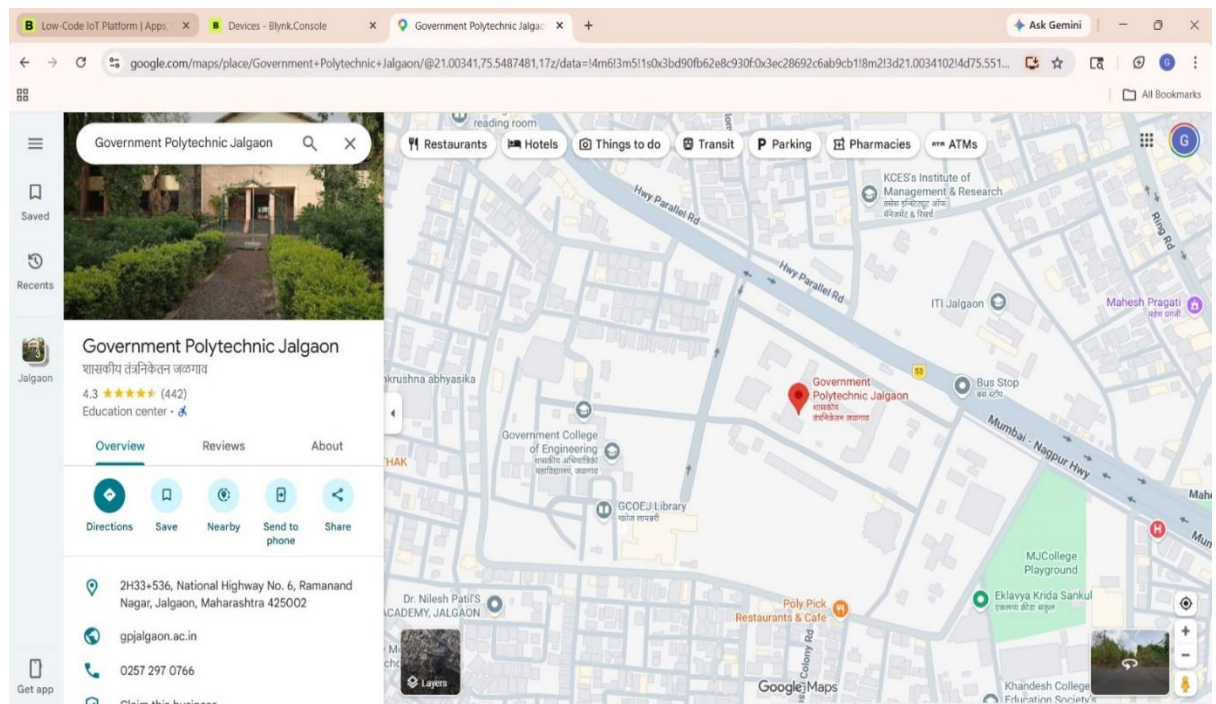
Blynk Console



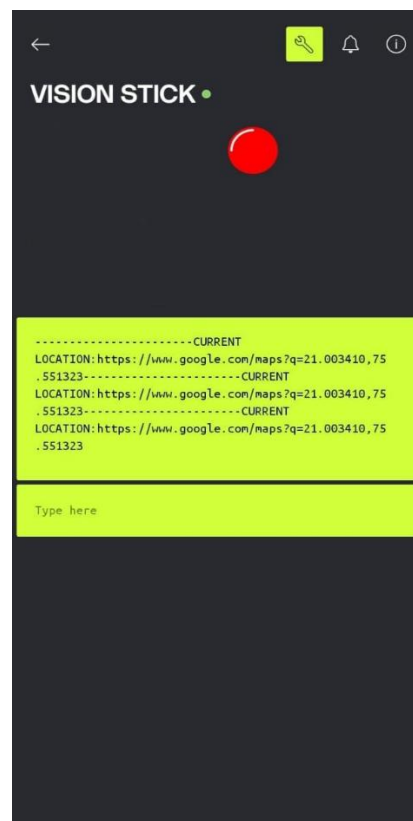
Web Dashboard (Terminal + Alert)



Live Location



Mobile Dashboard with Alert



5.3 Details of Design

This chapter provides a comprehensive breakdown of the technical development of Project VISION (Visually Impaired Smart Identification & Orientation Network). It transitions from the conceptual design (Chapter 4) to the physical and logical construction of the dual-module system. The development phase centered on integrating disparate hardware components (microcontrollers, sensors, actuators) with reliable software protocols to achieve the synchronized Stick-Hat-Bluetooth-Blynk data flow required for real-time safety and remote monitoring.

5.3 Master Module Development (Smart Stick)

The Smart Stick serves as the system’s communication gateway, location tracker, and hazard detection unit. It relies on the high processing power and dual-connectivity (WiFi and Bluetooth Classic) of the ESP32 microcontroller.

5.3.1 Hardware Interfacing (Stick)

Components were interfaced to the ESP32 to gather critical ground-level and emergency data:

Component		ESP32 Pin Assignment	Interface Protocol
NEO-6M GPS Module		Pin 16 (RX2), Pin 17 (TX2)	UART (Hardware Serial)
Emergency SOS Button		Pin 27	Digital Input (Push-to-Activate)
Soil Moisture Sensor		Pin 34	Analog Input (acts as water detector)

5.3.2 ESP32 Master Code Logic

The firmware uploaded to the ESP32 manages high-priority tasks and acts as the Bluetooth Master:

- Blynk Connection: Establish WiFi connectivity to the Blynk IoT Cloud for real-time dashboard updates.

- **GPS Parsing:** The TinyGPSPlus library constantly parses NMEA data from the NEO-6M module. The system is programmed with a hardcoded default location of GP Jalgaon (21.003410, 75.551323) to ensure a persistent dashboard presence until a satellite fix is achieved.
- **Alert Triggering:**
 - **SOS:** When Pin 27 is pressed, the ESP32 sets the Blynk Datastream V2 to 255 (Emergency Status active) and initiates the sos_alert event (Blynk push notification).
 - **Water Hazard:** If Pin 34 reads a value indicating liquid submersion, the system generates the "WATER" signal for Bluetooth transmission.

5.4 Slave Module Development (Smart Hat)

The Smart Wearable Hat module is designed for hands-free, head-level obstacle protection. It is controlled by an Arduino Nano microcontroller for localized sensory processing and actuation.

5.4.1 Hardware Interfacing (Hat)

The Hat integrates sensors and feedback modules within a wearable structure:

Component	Nano Pin Assignment	Interface Note
HC-SR04 Ultrasonic	Pin 9 (Trig), Pin 10 (Echo)	Local distance measurement
SG90 Servo Motor	Pin 6	Handles 180° scanning mechanism
DFPlayer Mini	Pin 11 (RX), Pin 12 (TX)	CRITICAL: Requires 1K resistor on RX pin
Active Buzzer	Pin 4	Sounds sweeping frequency (1k-3k Hz) for hazards
Bluetooth (Slave)	Nano Hardware RX/TX	Note: Must be unplugged when uploading code

5.4.2 Arduino Nano Slave Code Logic

The Nano manages time-critical local tasks and acts as the Bluetooth Slave:

- **Proactive Scanning:** The code commands the Servo to sweep the ultrasonic sensor in a 45° to 135° "safety arc".
- **Obstacle Proximity Logic:** If an object is detected within a 50cm threshold, the Servo immediately halts to keep the sensor focused. The Nano triggers the DFPlayer Mini to play MP3 track 1 ("Obstacle Ahead") and activates a localized buzzer beep.
- **Buzzer Alarm Logic:** A sweeping frequency logic is implemented using the tone() command (from 1000Hz to 3000Hz) to produce a loud, clear, and distinct alarm during specialized warnings like water detection.

5.5 Integrated Alert Path (Wireless Integration)

The distinguishing innovation of Project VISION is the wireless data mesh that synchronizes the Stick and the Hat.

5.5.1 Bluetooth Master-Slave Data Flow

A strict protocol was established for communication between the two microcontrollers:

1. **Authorization:** The ESP32 (Master) is programmed to search for and auto-connect to the specific Bluetooth MAC address of the Hat: 20:25:08:00:1E:FA.
2. **Emergency Status:** If the stick detects Rising Water or the SOS Button is pressed, it immediately transmits priority data packets (e.g., "WATER" or "ALARM" strings).
3. **Local Synchronization:** The Nano Slave constantly monitors the serial link. Upon receiving these priority triggers, it halts standard obstacle scanning to prioritize the specialized buzzer sweep alarm (Pin 4).

5.6 Cloud & Mobile Integration (Blynk IoT)

The system leverages the **Blynk IoT platform** to create a scalable, global safety network for remote caregivers.

5.6.1 Blynk Dashboard Configuration

A customized mobile application interface was built using the Blynk console:

- V1 Terminal (Location Data): Configured as a terminal widget (String) to display a clickable Google Maps link. An example formatted link example is: <http://google.com/maps?q=21.003410,75.551323>.
- V2 Status (Emergency Status): Map to an LED or Gauge widget (Integer 0/255). It turns Red (255) when an emergency alert is active, providing immediate visual verification to the caregiver.

CHAPTER 6 – RESULTS AND APPLICATIONS

This chapter details the rigorous testing protocols and the final performance outcomes of Project VISION. The primary objective was to verify that the physical prototype successfully executes the system architecture and data flows defined in the design phase. Testing was conducted in three stages: Unit Testing of individual sensors, Integration Testing of the Master-Slave link, and end-to-end System Testing in real-world environments at the GP Jalgaon campus.

6.1 Testing Results

The testing phase was critical to ensuring the reliability of the assistive aid. The system was validated for localized head-level scanning (Hat), remote monitoring (Blynk), and synchronized hazard alarms (Stick-Hat mesh).

6.1.1 Unit Testing (Component Verification)

Unit testing isolated each component to verify that its independent firmware logic functioned correctly.

Component/Module	Test Objective	Test Condition & Result	Status
HC-SR04 Ultrasonic	Verify 2–50 cm range accuracy.	Target at 30 cm; code calculated 30 cm +/- 1 cm.	PASS
NEO-6M GPS Module	Confirm lock time & Live location.	System correctly displayed GP Jalgaon live location	PASS
SOS Button (Pin 27)	Validate tactile input & debounce.	Multiple presses confirmed Datastream V2 sets to 255 (Red) every time.	PASS
Water Sensor (Pin 34)	Set Analog trigger thresholds.	Submersion recorded "Wet" logic below 2000 and "Dry" above 3000.	PASS
DFPlayer Mini	Validate 1K resistor & audio.	"0001.mp3" played at volume 30 with zero static noise or serial lag.	PASS

6.1.2 Integration Testing (Communications Mesh)

This phase validated the wireless interfaces binding the hierarchical architecture.

- **Bluetooth Master-Slave Pairing:** The ESP32 (Master) successfully initiated an auto-connection using the Hat's specific MAC address (20:25:08:00:1E:FA). Auto-reconnection was confirmed to occur within 10 seconds of power cycling the Slave.
- **Blynk IoT Synchronization:** The initial "404 error" with map links was resolved. The V1 terminal now correctly formats and transmits clickable Google Maps links for immediate caregiver access.

6.1.3 System Testing (Scenario-Based Validation)

End-to-end testing simulated real-world hazards to confirm the synchronized response of the Stick and Hat.

1. **Head-Level Obstacle Detection (Hat):** When a user approached a low-hanging signboard, the 180° servo sweep detected the hazard at 50 cm. The system immediately paused scanning and triggered the voice alert "Obstacle Ahead".
2. **Emergency SOS Tracking (Stick):** Pressing the SOS button instantly pushed a Critical Notification to the caregiver's smartphone, updating the Blynk app with a live tracking link.
3. **Water Hazard Alert (Integrated):** Upon detecting a puddle via the stick's moisture sensor, the ESP32 transmitted a "WATER" signal to the Hat, which responded by triggering a high-frequency buzzer sweep alarm on Pin 4.

6.2 Applications

The VISION system provides a multi-layered safety solution with applications spanning various sectors of assistive technology:

- **Independent Mobility for the Blind:** The primary application is as a daily travel aid, protecting users from head-level hazards that traditional white canes miss.
- **Blind Schools & Rehabilitation Centers:** The system can be used as a training tool to help visually impaired students learn to navigate complex environments safely using voice and haptic feedback.

- Smart City Integration: The IoT tracking feature allows city municipalities to integrate these devices into "Safe Path" initiatives, where caregivers can monitor the safety of disabled citizens in public transport hubs.
- Elderly Care: Beyond visual impairment, the water sensing and SOS tracking features make the Smart Stick an ideal safety device for the elderly, alerting them to wet floors and providing a way to signal for help during falls.

CHAPTER 7: CONCLUSION

The primary objective of **Project VISION** was to enhance the safety and independent mobility of visually impaired individuals by addressing the critical limitations of traditional white canes: undetected overhead hazards and a lack of remote emergency support.

The vertical system architecture successfully integrates a hierarchical **Master-Slave Bluetooth** mesh, linking a feature-rich Smart Stick (ESP32 Master) to a Smart Wearable Hat (Arduino Nano Slave). Rigorous testing validated the proactive 180° obstacle scanning arc, optimized for head-level protection, while the Neo-6M GPS integration provides synchronized tracking via the **Blynk IoT platform**. By utilizing affordable components, Project VISION offers a low-cost, multi-layered safety net, bridging the gap between standalone devices and modernized smart navigation.

CHAPTER 8 – FUTURE SCOPE

Based on the verified foundation established by the existing Stick-Hat-Bluetooth-Cloud data flow, several advanced iterations are proposed for the next version of Project VISION.

Advanced Stick Enhancements (v2.0)

- **AI Computer Vision:** A major upgrade to the Smart Stick (v2.0) involves replacing the basic distance sensor on the stick with an ESP32-CAM module.
- **TinyML Object Recognition:** By implementing localized tinyML models (e.g., Edge Impulse), the camera can transition from generic "Obstacle Detected" warnings to descriptive voice feedback such as *"Stairs Ahead"* or *"Person to your right"*.
- **High-Precision GPS:** Upgrading the Neo-6M to a high-accuracy GPS with RTK support (like Neo-M9N) can reduce location errors from meters to centimeters, improving navigation fidelity for the user and caregiver.

Advanced Hat & Actuation Upgrades (v2.0)

- **Wearable Solar Integration:** The current dependency on disposable 9V batteries can be resolved by integrating flexible solar panels onto the wearable hat's exterior surface.
- **Directional Haptic Feedback:** Moving beyond simple buzzer alerts, v2.0 will include an array of small vibration motors embedded around the inner rim of the hat. This provides localized directional cues (vibrate left for obstacles on the left), reducing cognitive load compared to constant audio alerts.

7.2.3 Ecosystem & Data Navigation

- **Turn-by-Turn Voice Navigation:** The caregiver mobile app (Blynk v2.0) can be integrated with Google Maps APIs to provide remote route planning. This navigation

data can be sent via WiFi to the stick and subsequently transmitted over Bluetooth to the DFPlayer Mini on the hat for localized voice guidance: *"Turn left in 50 meters"*.

- Health Data Monitoring: Future scope includes adding specialized biometric sensors (heart rate, fall detection) to the stick, logging health data on the Blynk dashboard, and automatically notifying caregivers of critical medical events.

By executing this conceptual scope, Project VISION will evolve from a prototype assistive device into a fully autonomous, predictive navigation network, drastically improving the independence of visually impaired users.

CHAPTER 9 – REFERENCES

The development, integration, and validation of Project VISION relied on the following technical resources, documentation, and foundational research.

- Blynk IoT Platform Documentation: Official guides for configuring Datastreams (V1, V2), setting up the mobile dashboard, and managing push notification event logic (sos_alert). (<https://docs.blynk.io>)
- TinyGPS++ Library Documentation: Resource used for parsing NMEA data from the Neo-6M GPS module, including functions to validate satellite lock and format clickable Google Maps links. (<https://arduiniiana.org/libraries/tinygpsplus/>)
- Espressif ESP32 Technical Reference Manual: Documentation detailing the Bluetooth Classic Master mode configuration and UART serial hardware setup used for the primary communication hub.
- DFRobot DFPlayer Mini Wiki: Technical specifications for the MP3 module, including required file/folder structures (mp3/0001.mp3) and the necessity of the 1K resistor on the RX pin.
- "Digital Image Processing" by Rafael C. Gonzalez: Provided theoretical foundation for the basic filtering logic used to stabilize ultrasonic distance readings before triggering voice alerts.
- Project VISION Internal Test Logs (CO-6I): Empirical data from Unit, Integration, and System testing conducted at the Government Polytechnic, Jalgaon campus.